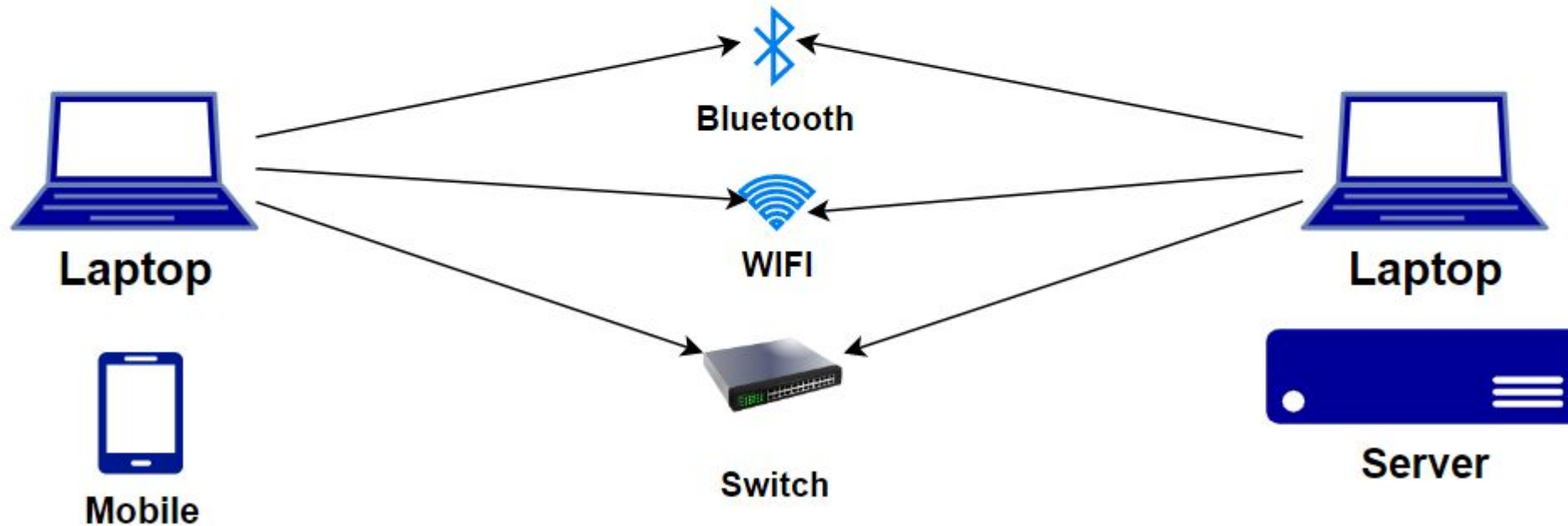


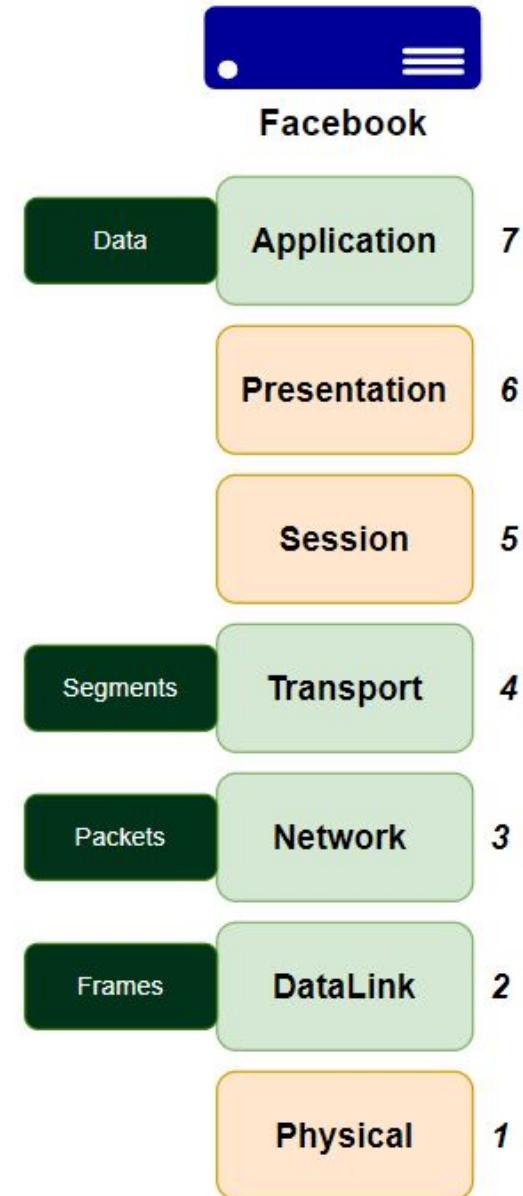
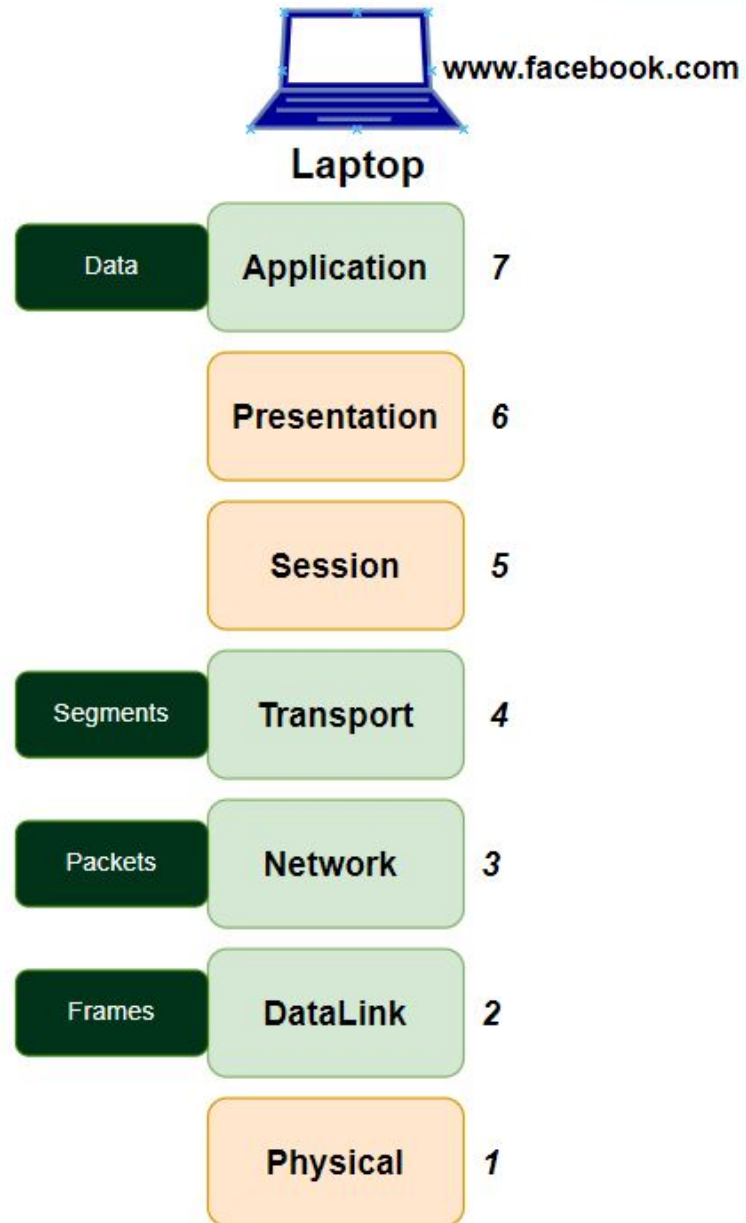
What is a network?

A network is

- a group of devices (Laptop / Mobile / Server)
- using specific medium (WIFI / Bluetooth / Cables/ Router / Switch)
- to deliver specific service (File Transfer / Browsing)



OSI MODEL



Application Layer like HTTP (Hyper text transfer protocol)
FTP (File Transfer Protocol)

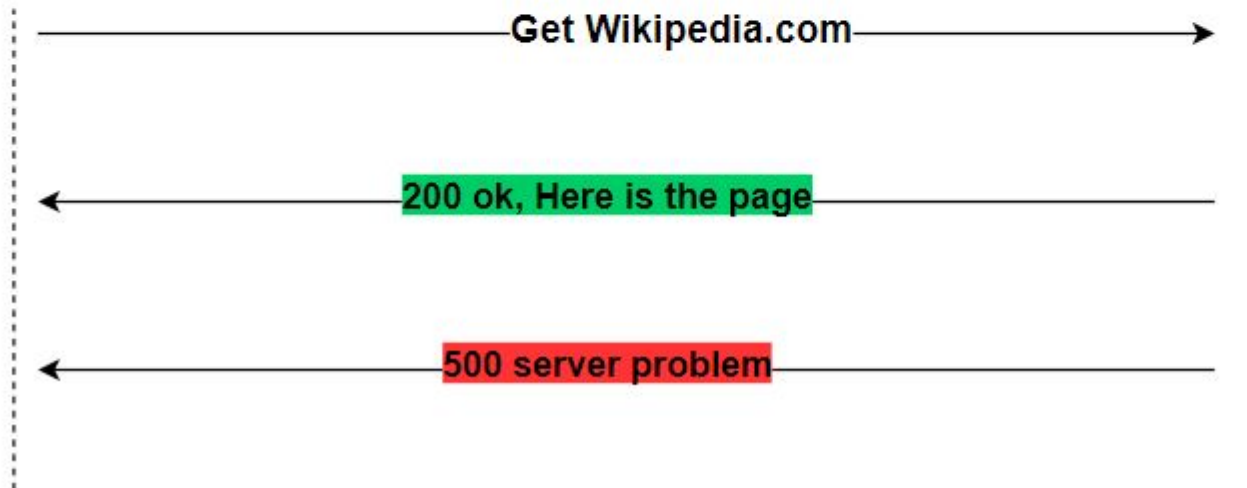
A Protocol is a language or method of communicating between client & Server to provide service
like browsing or file transfer



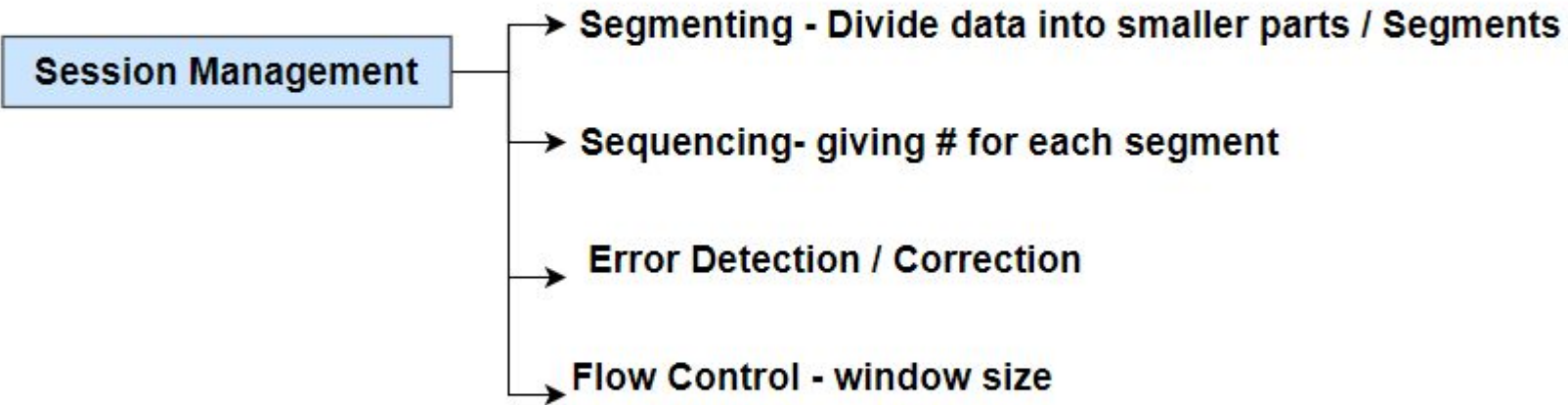
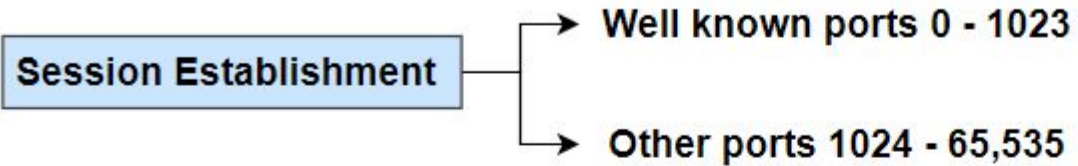
Laptop



WikiPedia



Transport Layer has two protocols TCP or UDP



Session Termination

TCP

HTTP > 80
HTTPS > 443
SSH > 22

UDP

DNS > 53

TCP Protocol

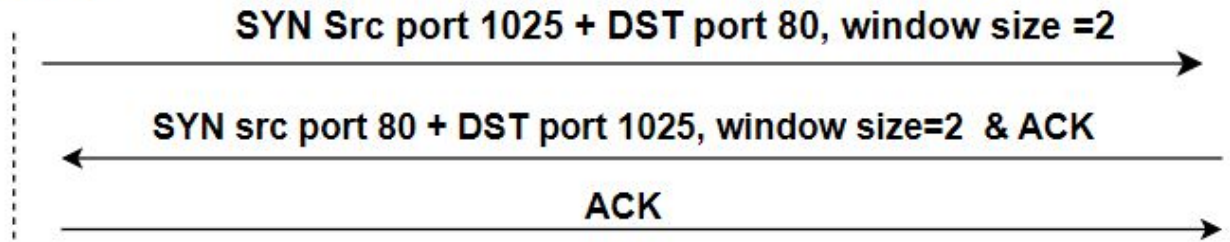


Laptop

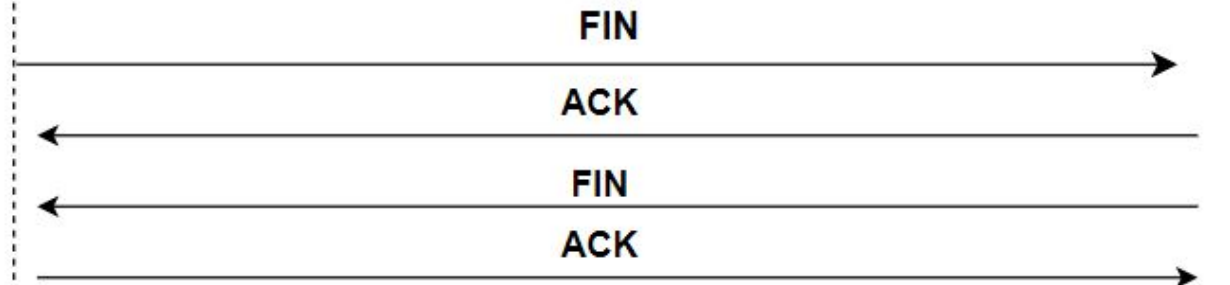
TCP - 3 way handshake



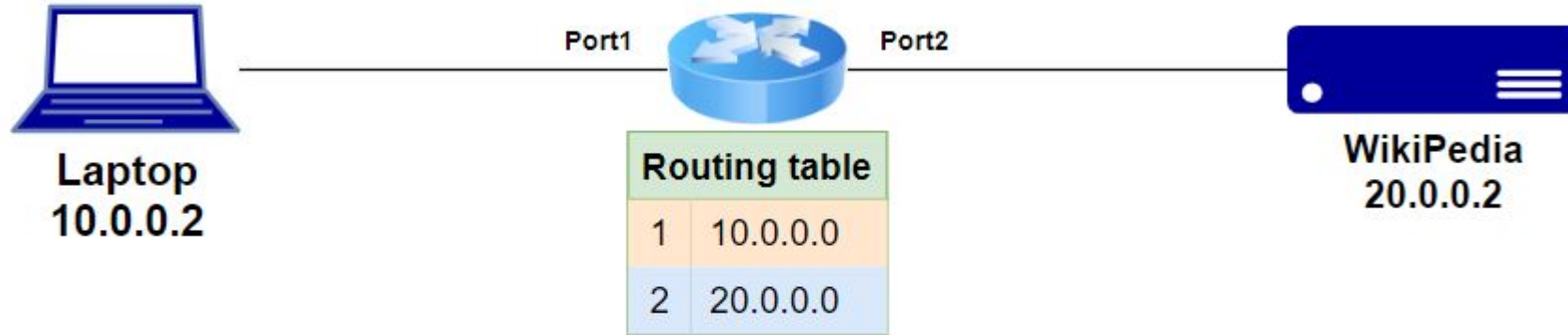
WikiPedia



TCP - 4 way handshake



Network Layer used for end to end delivery of packet in different networks



Routed Protocol

Logical address assigned to each device

IPv4

IPv6

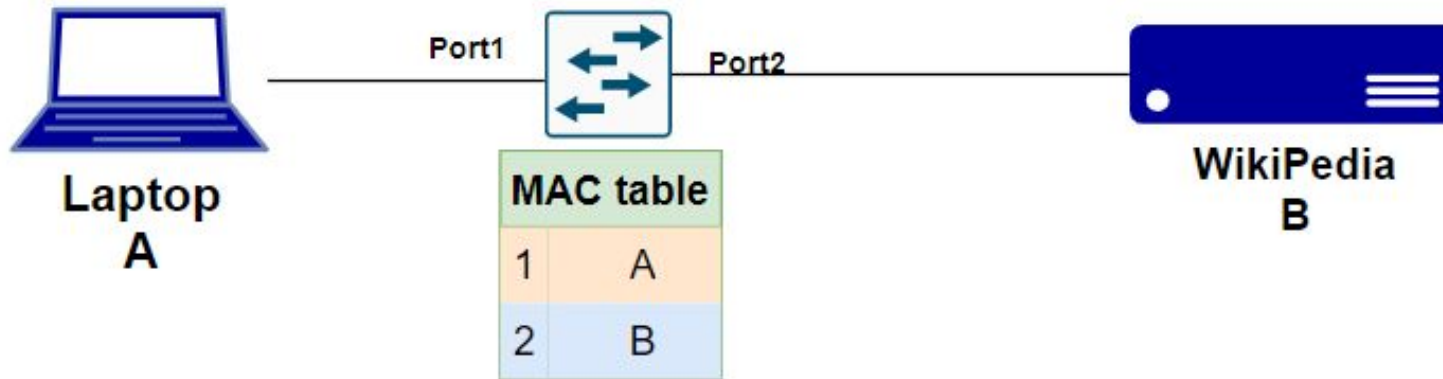
Routing Protocol

Build routing table to choose best path to reach destination IP

OSPF

BGP

Datalink Layer used for hop to hop delivery of the frame in the same network

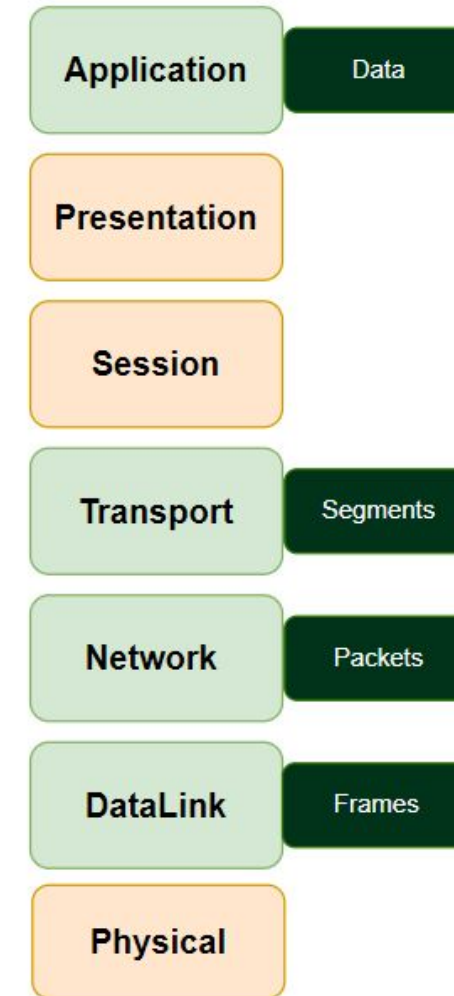
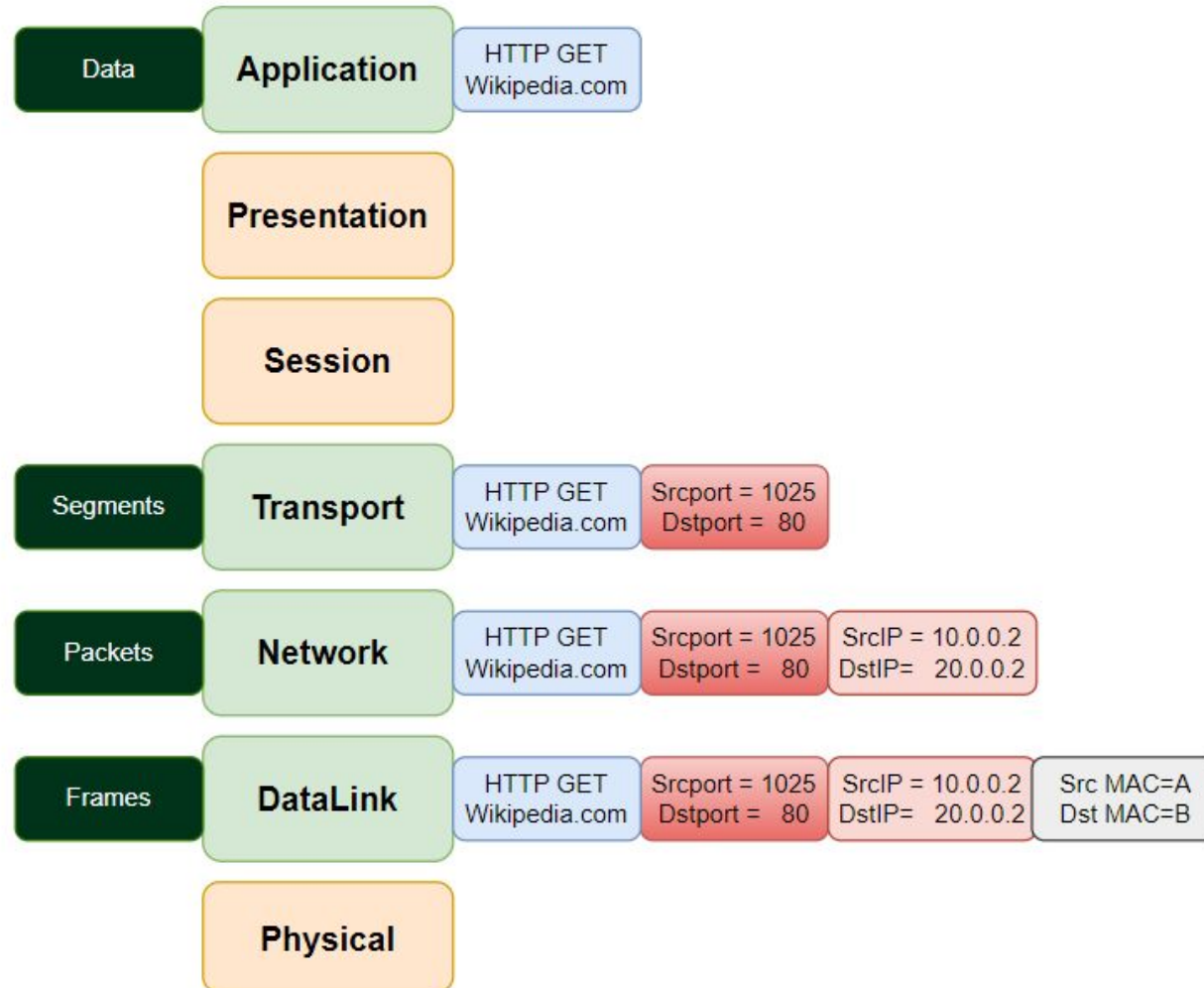
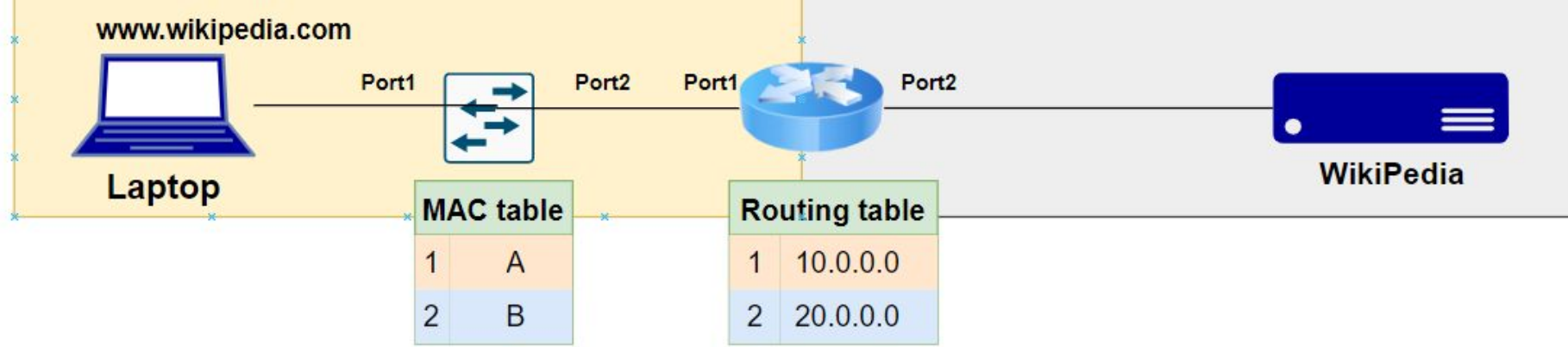


Mac address

Layer 2 / physical address

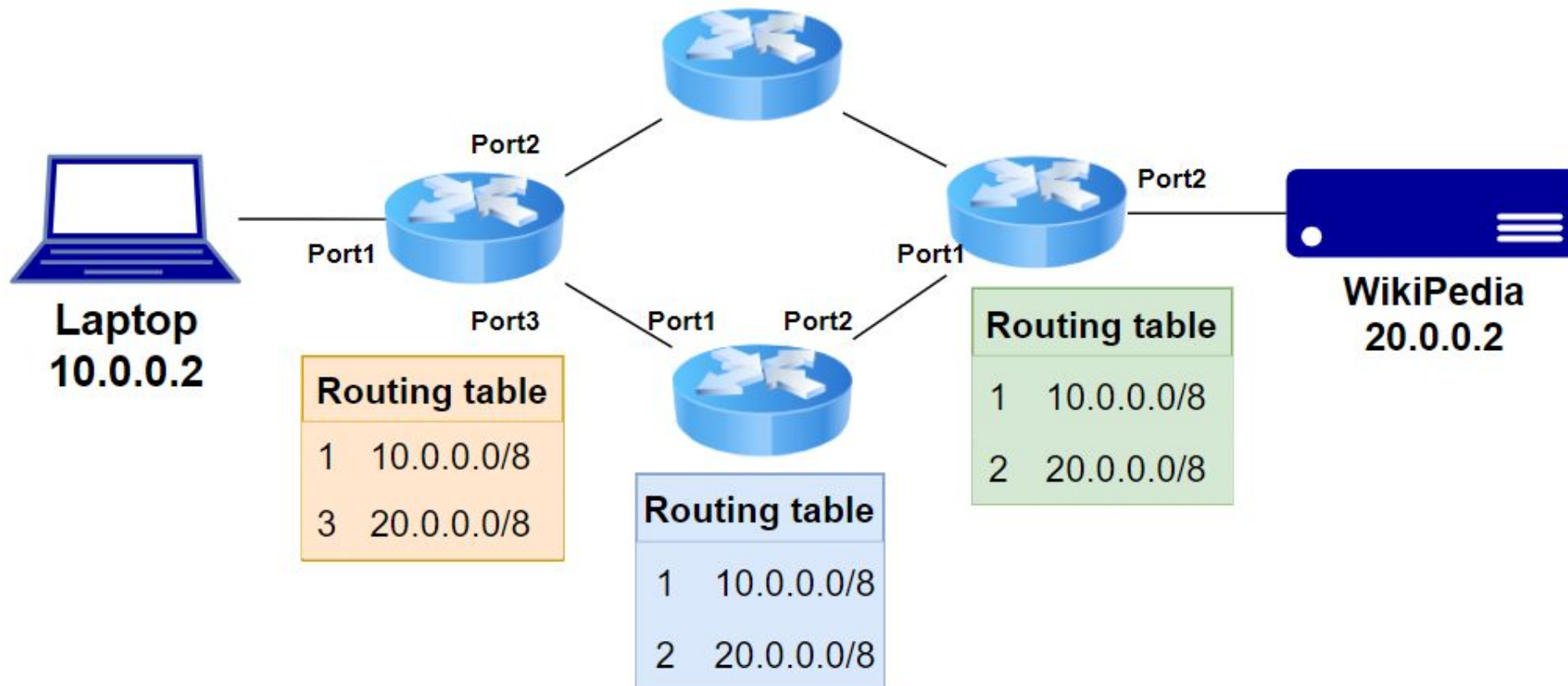
Mac Table

Switch use it to forward frame based on
MAC address



IPv4

- Gives layer 3 address - End to End address for computer/servers
- Used as a Network address in the Routing table of every Router
- Network address always comes with subnet mask to define The Network part & The Host part



IPv4

- 32 bits address - Represented in decimal or binary
- 4 octets, each octet is from 0 - 255

172.255.0.15

128 64 32 16 8 4 2 1

10101100 .11111111. 00000000. 00001111

Subnet mask

- Defines the network part and host part
- varies from 0 - 32
- can be represented in 4 octets, each octet 0 - 255

128 64 32 16 8 4 2 1

/8

11111111.00000000.00000000.00000000

255.0.0.0

Class A 1 - 127

/16

11111111.11111111.00000000.00000000

255.255.0.0

Class B 128 - 191

/24

11111111.11111111.11111111.00000000

255.255.255.0

Class C 192- 223

/26

11111111.11111111.11111111.11000000

255.255.255.192

Network address

- Use by Routing table to choose the best path
- Its part in the IP knows from the 1's of the subnet mask

Host address

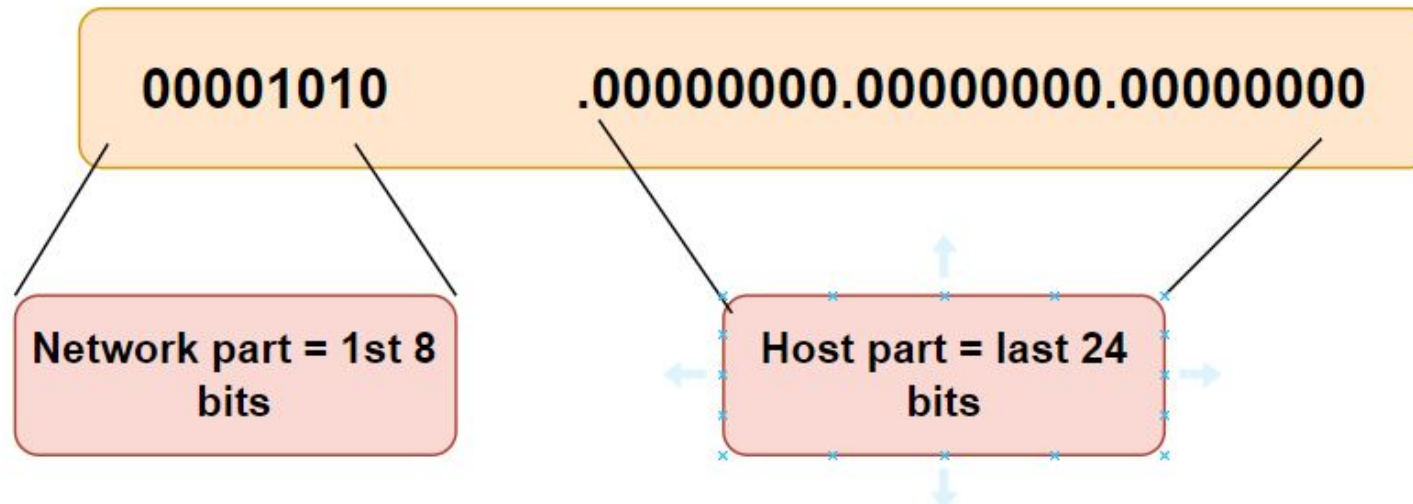
- Used by the Computers / laptops / server
- Its part is known from the zeros on subnet mask

Broadcast address

- Extra address , Used by all the machines

10.0.0.0 /8

128 64 32 16 8 4 2 1



Exercise1

10.0.1.5 /24

128 64 32 16 8 4 2 1

00001010 .00000000. 00000001. 00000101

11111111 .11111111. 11111111. 00000000

255.255.255.0

Network address

10.0.1.0

of IPs

256

of Host

254

Broadcast address

10.0.1.255/24

Subnet

- Slicing one network into # of networks called subnets

Exercise2

- Slice network 10.0.0.0/24 to be four networks
- how many hosts in each subnet
- what subnet address
- host address
- broadcast address

10.0.0.0 /24

128 64 32 16 8 4 2 1

00001010 .00000000. 00000000. 00000000

of subnets

of IPs

10.0.0.0 /24

00001010 .00000000. 00000000. 00 000000

Subnet1 address

Subnet2 address

Subnet3 address

Subnet4 address

BC1 address

BC2 address

BC3 address

BC4 address



IPv4

- $2^{32} = 4,294,967,296$ IP available only
- We have shortage IPs since each device needs a unique IP

Private IP addresses

- Private IP addresses are not routable on the Internet.
- Are used for internal communication within a network.

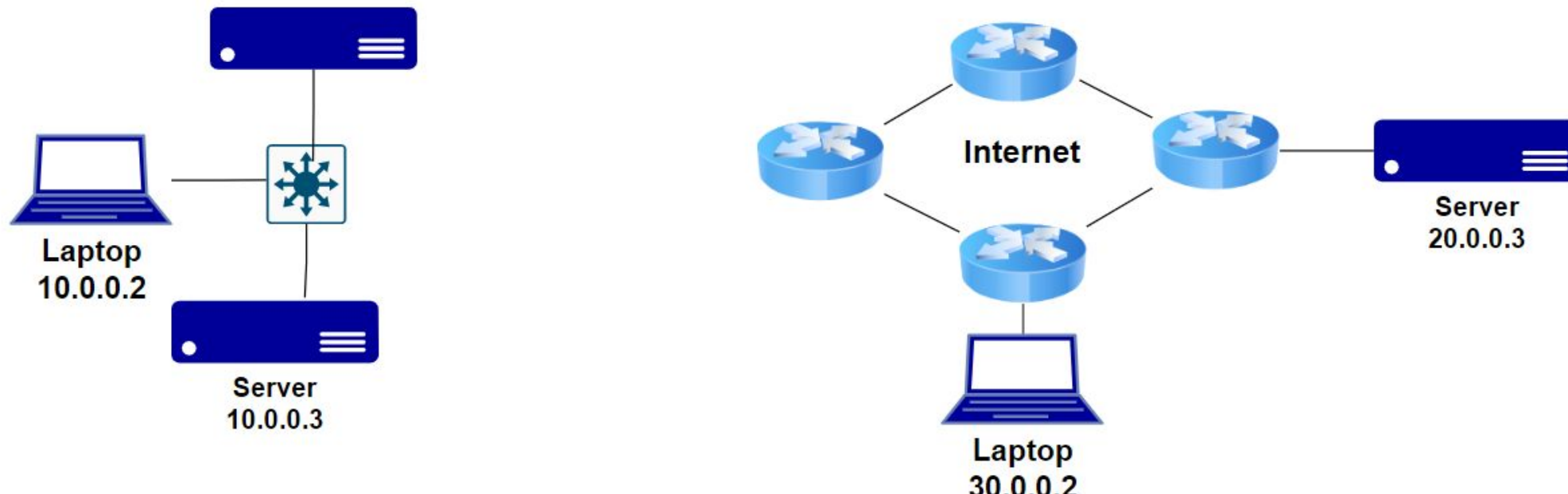
10.0.0.0 - 10.255.255.255 (10.0.0.0/8)

172.16.0.0 - 172.31.255.255 (172.16.0.0/12)

192.168.0.0 - 192.168.255.255 (192.168.0.0/16)

Public IP addresses

- These are globally unique IP addresses assigned to devices connected to the Internet.
- Public IP addresses are allocated and managed by IANA

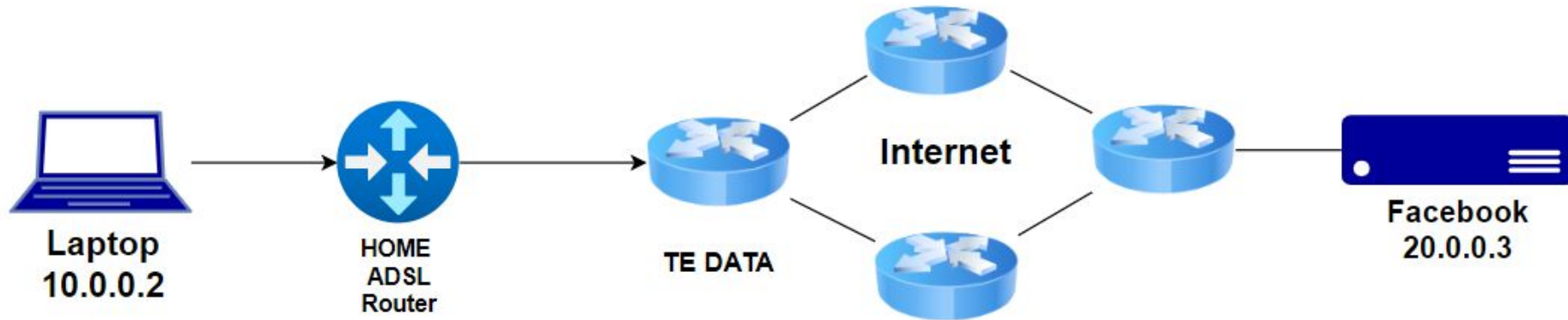


CMD + What is my ip to show private and public

Next episode I will show you how this is happening

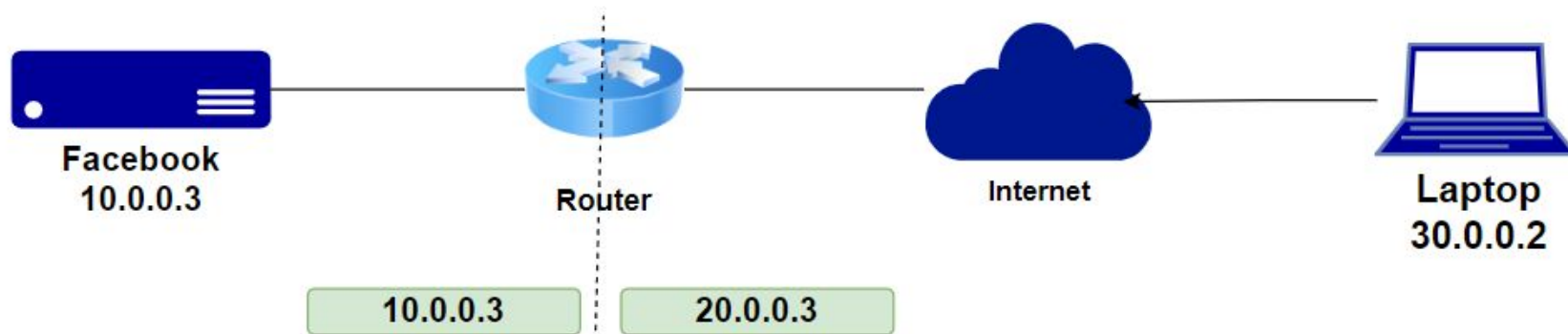
NAT

- Network Address Translation.
- Router or any layer 3 device (firewall) changes private IP to a public IP in the packet headers to make it routable over internet.



Static NAT

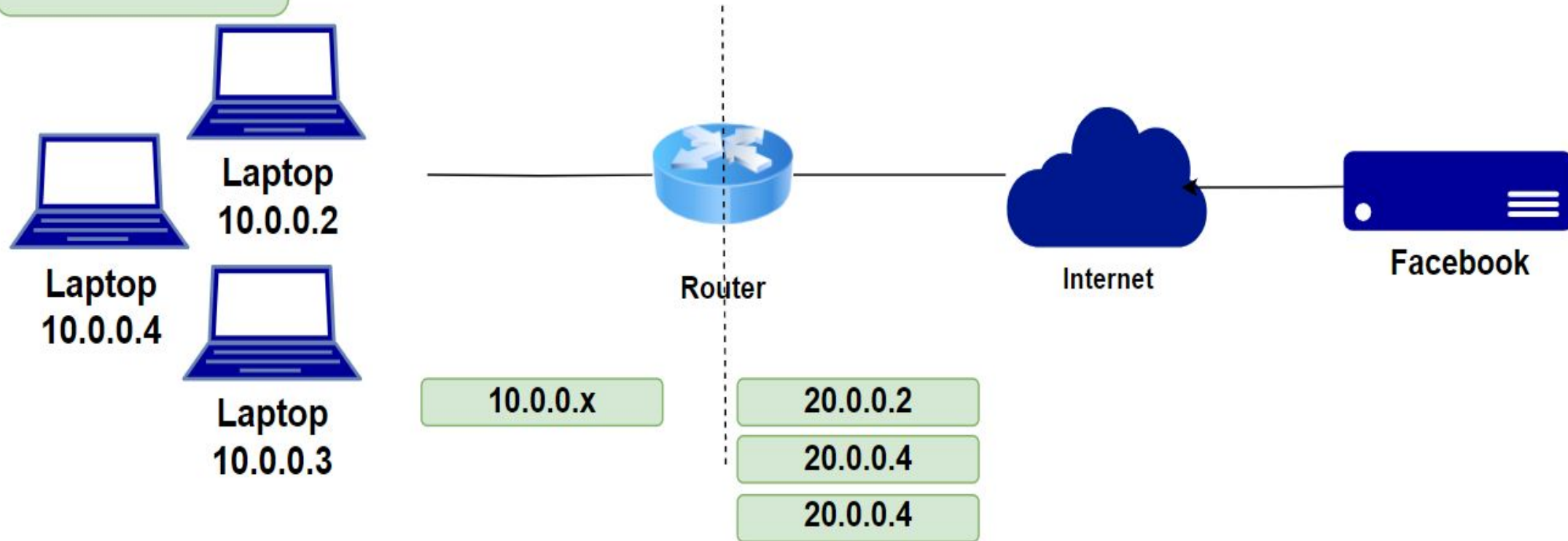
- One to one mapping, One private IP to one public IP
- Used when a device inside needs to be accessible from the internet, like a web server.



Inside Local	Inside Global
10.0.0.3	20.0.0.3

Dynamic NAT

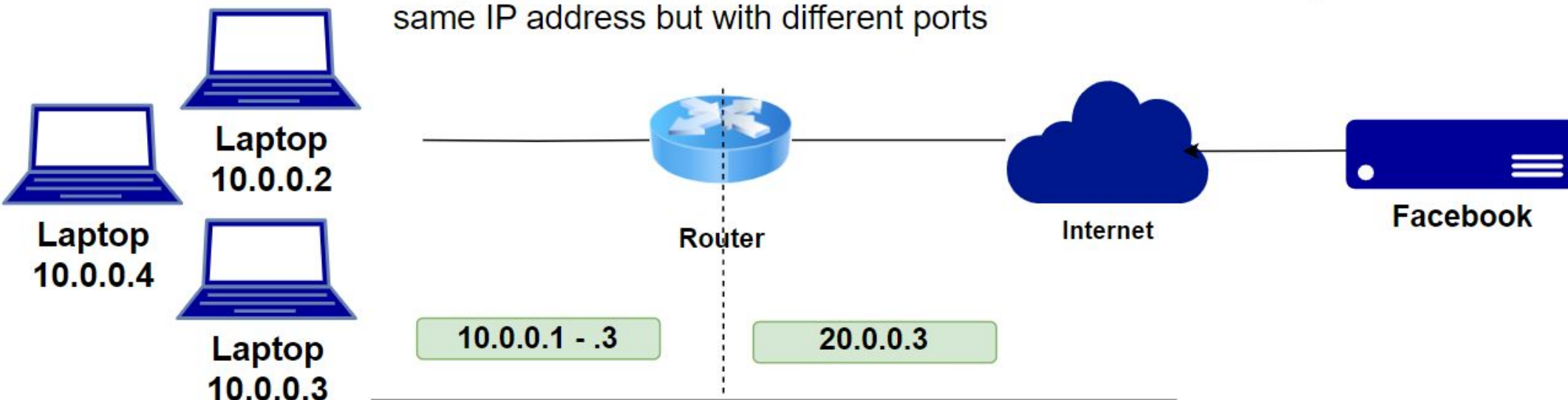
- Dynamic NAT has a pool of public IP addresses and assigns them to private devices.



Inside Local	Inside Global
10.0.0.2	20.0.0.2
10.0.0.3	20.0.0.3
10.0.0.4	20.0.0.4

PAT

- Port address Translation
- Many devices on a local network can access the internet using the same IP address but with different ports



Inside Local	Inside Global
10.0.0.2:10001	20.0.0.3:40001
10.0.0.2:10002	20.0.0.3:40002
10.0.0.3:10001	20.0.0.3:40003
10.0.0.3:10002	20.0.0.3:40004
10.0.0.4:10001	20.0.0.3:40005

Router

- Layer 3 device, check only IP
- Connect different networks
- Each interface / port is in a different network
- Each port takes IP in its network
- Uses Routing Table to route the packets



10.0.0.0/24

20.0.0.0/24

Port1
10.0.01

Port2
20.0.01

Laptop
10.0.0.2

Server
20.0.0.2

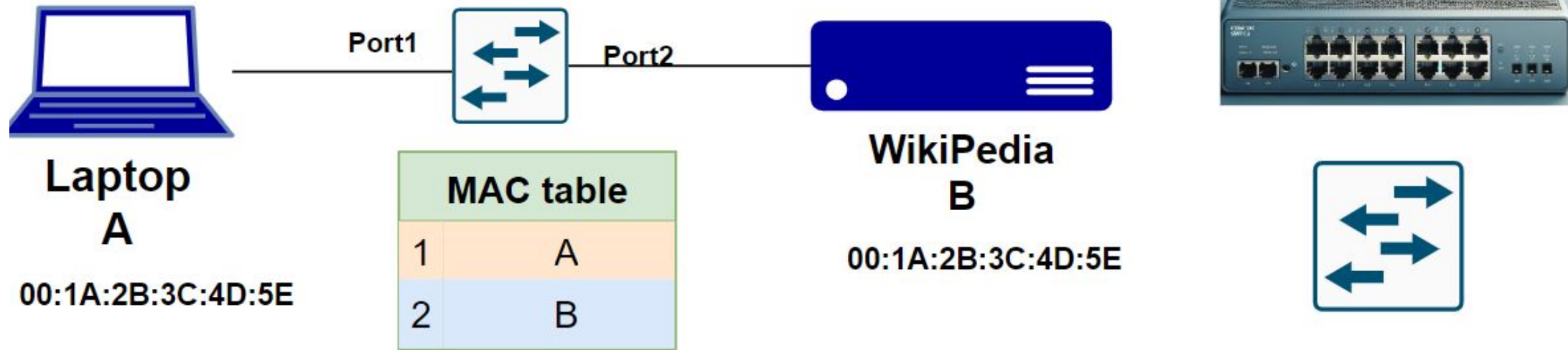
Router

Routing table

1	10.0.0.0/24
2	20.0.0.0/24

Switch

- Connect devices in the same layer 2 technology
- Now days this technology will be only Ethernet
- Ethernet uses MAC address and switch will use MAC Table to forward the frames

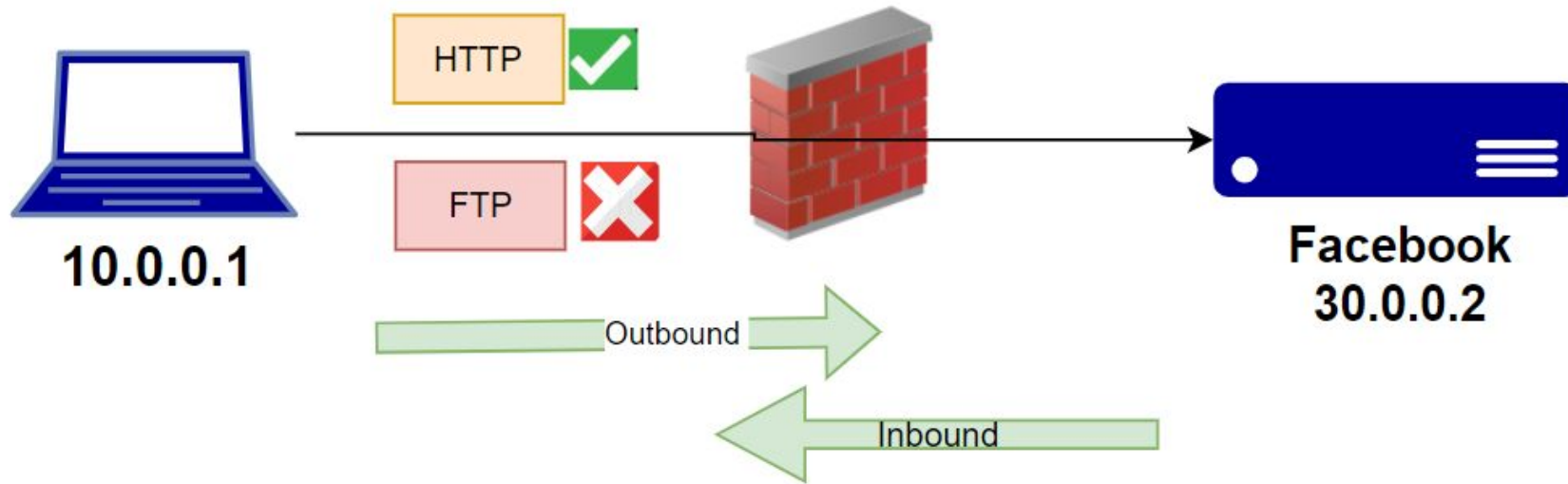


Multi layer switch MLS

- a Layer 3 switch, operates at both the Data Link layer (Layer 2) and the Network layer (Layer 3)
- Like switches, multilayer switches can forward packets based on MAC addresses
- like a router forwarding Packets based on IP addresses

Firewall

- Protect network by allow specific IPs on specific ports
- Allowed traffic **statefully** that is initiated outbound or inbound
- Works mainly on Layer3 & Layer4 &
- NGFW works on layer 7



Intrusion Prevention System (IPS)

- **IDS** Inspecting network packets **Detects only & generate alarms** while **IPS** blocks and prevent attacks
- **Signature-based** detection, maintains a database of known attack signatures, If the IPS detects a packet that matches a signature in its database, it will block it.

Intrusion Detection System (IDS)

- **Behavioral analysis** IPS monitors and analyzes network traffic over a period of time to understand typical patterns of communication, including the types of protocols used, traffic volume, and typical endpoints accessed. It looks for anomalies such as unusual traffic patterns, unexpected protocol usage.



Client



Web server

SRC IP

DST IP

SRC MAC

DST MAC

SRC MAC

on the Network Card

SRC IP

Static IP from windows settings

Internet Protocol Version 4 (TCP/IPv4) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address: 192 . 168 . 1 . 199

Subnet mask: 255 . 255 . 255 . 0

Default gateway: 192 . 168 . 1 . 1

SRC IP

Dynamic using DHCP



Client



DHCP server

DHCP Discovery

The client broadcasts a DHCPDISCOVER message on the network to find a DHCP server.

DHCP Offer

offering IP configuration information, IP 10.0.0.2/24 + GW 10.0.0.1 + DNS IP

DHCP Accept

DHCP ACK

DST IP

DNS - Domain name server



Client



DNS server

DNS Request

What is IP of google.com

DNS Reply

Google IP is 55.1.1.1

DST MAC

ARP - Address Resolution protocol -
Layer 2 protocol
Send on B.C MAC ffff.ffff.ffff



Client



server
10.0.0.4

ARP Request
What is MAC of IP 10.0.0.4

ARP Reply
MAC of IP 10.0.0.4 is C

Default Gateway



Laptop
10.0.0.2



10.0.0.1
B

ARP Request
What is MAC of IP 10.0.0.1

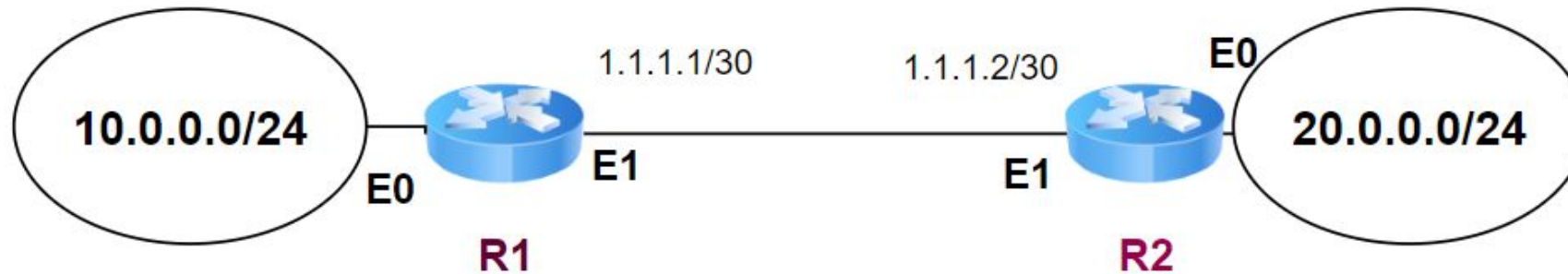
ARP Reply
MAC of IP 10.0.0.1 is B



Google

Static route

Used if the network is simple (few # of networks)
Admin distance 0 or 1



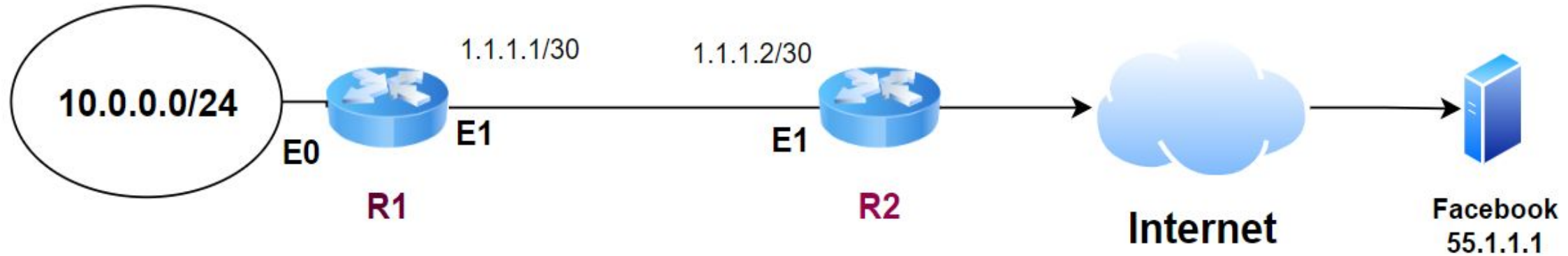
	Subnet	Interface	Next hop
C	10.0.0.0/24	E0	
C	1.1.1.1/30	E1	
S	20.0.0.0/24	E1	1.1.1.2

```
R1(Config)#ip router 20.0.0.0 255.255.255.0 E1  
R1(Config)#ip router 20.0.0.0 255.255.255.0 1.1.1.2
```

Default Route

When destination IP doesn't match any entry, send the traffic to the default route

```
R1(Config)#ip router 0.0.0.0 0.0.0.0 E1  
R1(Config)#ip router 0.0.0.0 0.0.0.0 1.1.1.2
```



	Subnet	Interface	Next hop
C	10.0.0.0/24	E0	
C	1.1.1.1/30	E1	
S*	0.0.0.0/0	E1	1.1.1.2

Admins Distance

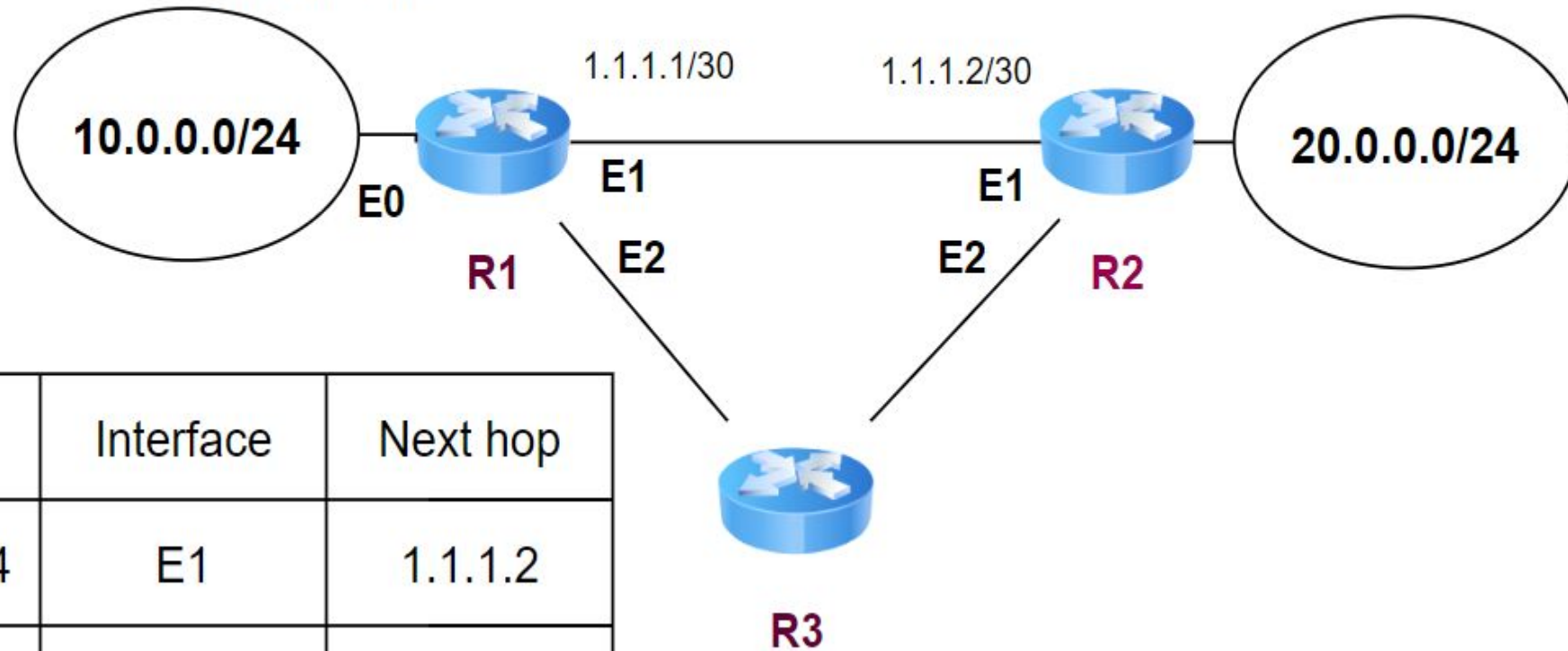
Give preference of some protocols over others - lower is better

Static - 0 or 1

OSPF - 110

iBGP - 200

EBGP - 20



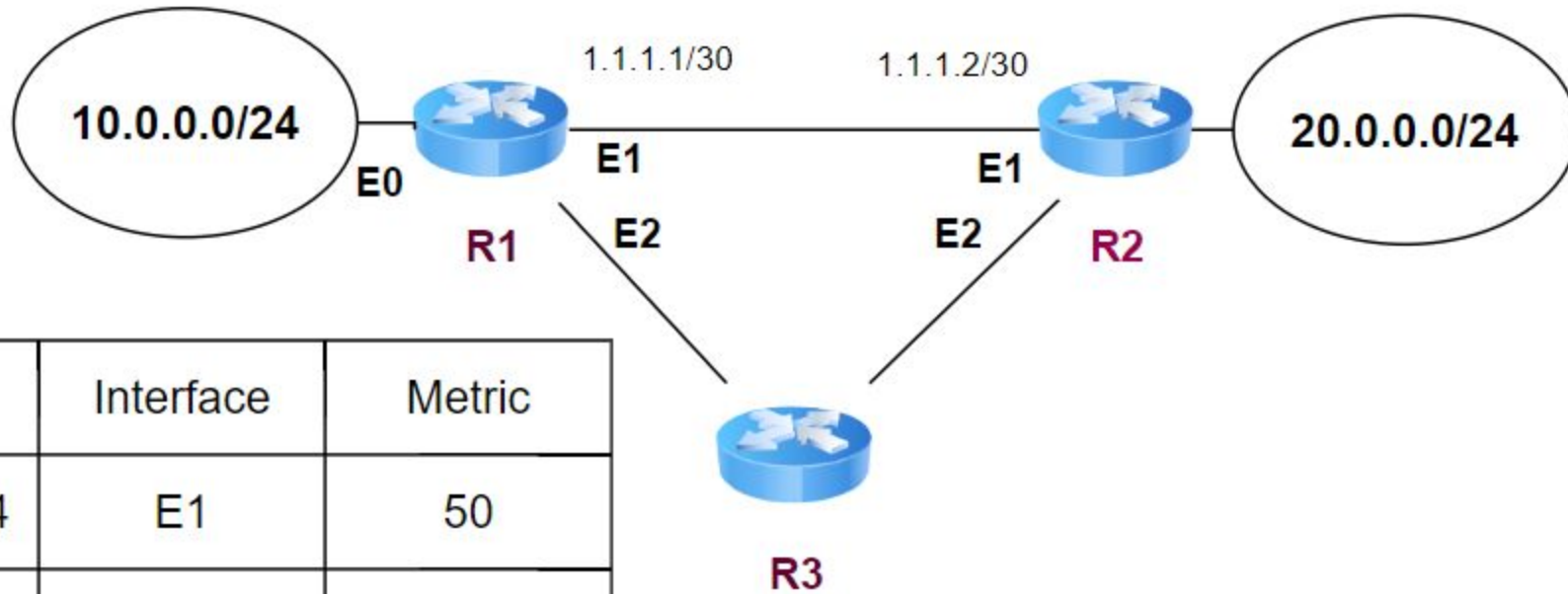
S

O

Subnet	Interface	Next hop
20.0.0.0/24	E1	1.1.1.2
20.0.0.0/24	E2	

Metric

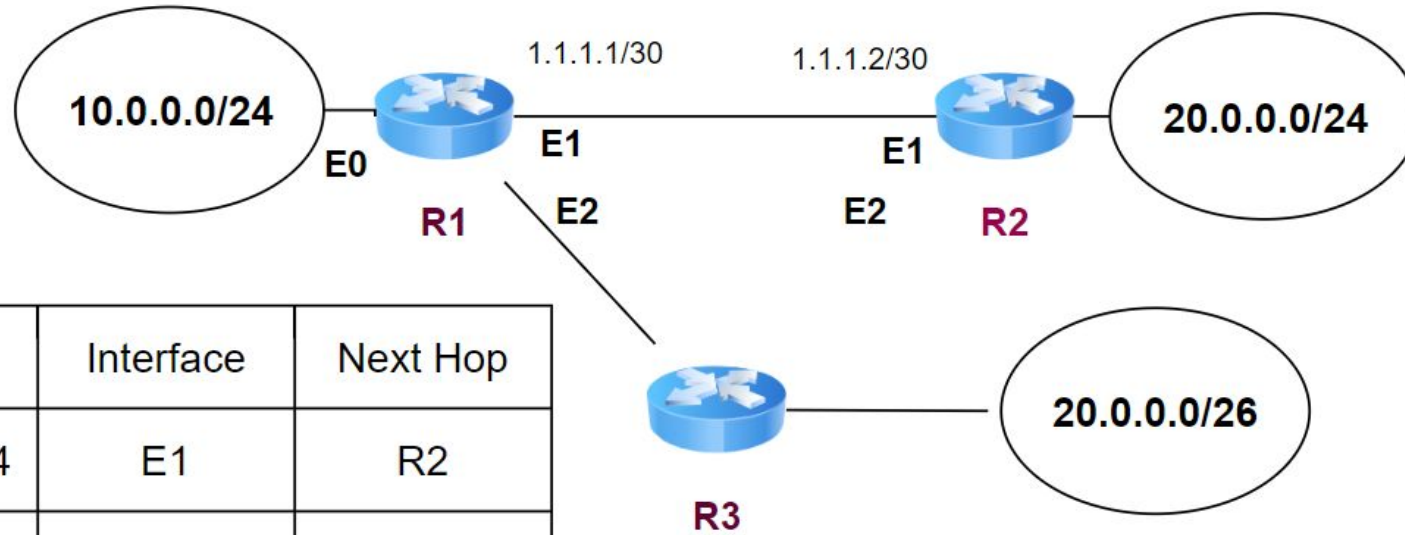
Cost to reach the destination
lower cost is better path (within the same protocol)



	Subnet	Interface	Metric
O	20.0.0.0/24	E1	50
O	20.0.0.0/24	E2	6-

Longest bit match

Preferred route is when Destination IP match the biggest network mask



S

O

Subnet	Interface	Next Hop
20.0.0.0/24	E1	R2
20.0.0.0/26	E2	R3

Best path selection

Longest bit match

Admins Distance

Metric

Ping

- Ping is a tool used to check the connection between two devices on a network.
- It sends a message called an ICMP (Internet Control Message Protocol) echo request & waits for a reply to see if the connection is good.

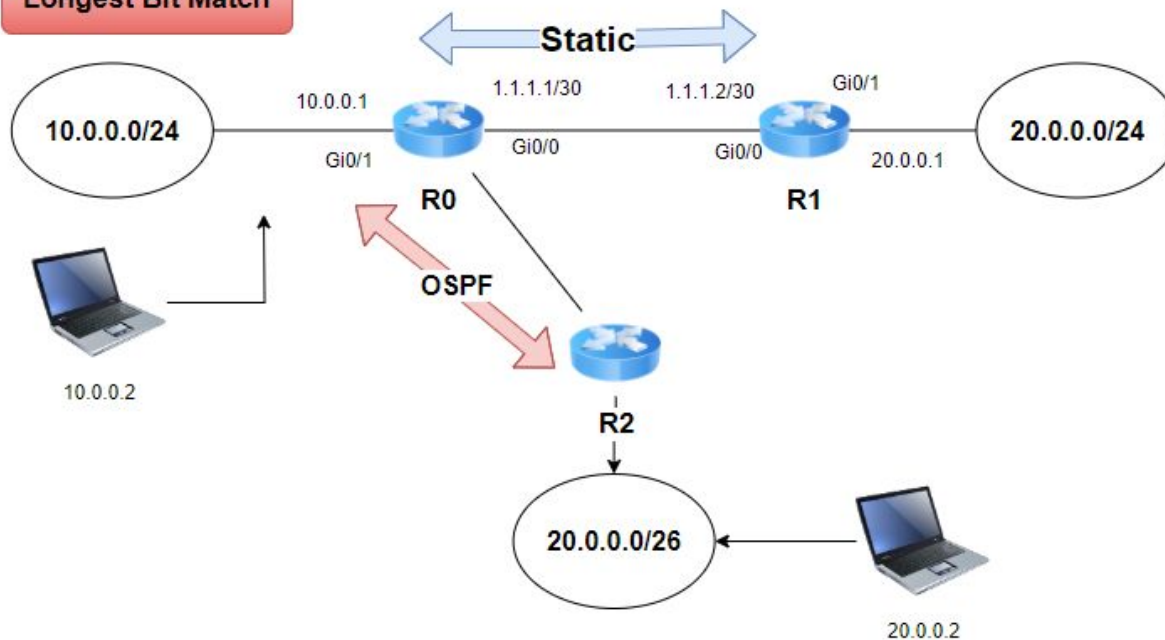
TTL

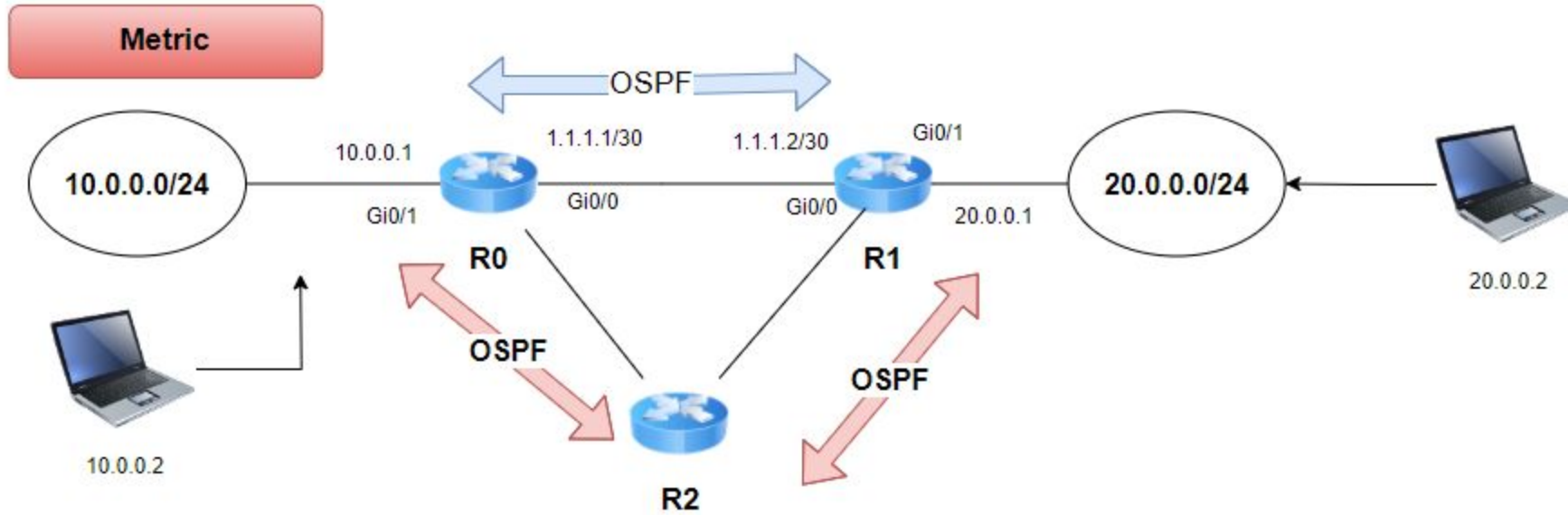
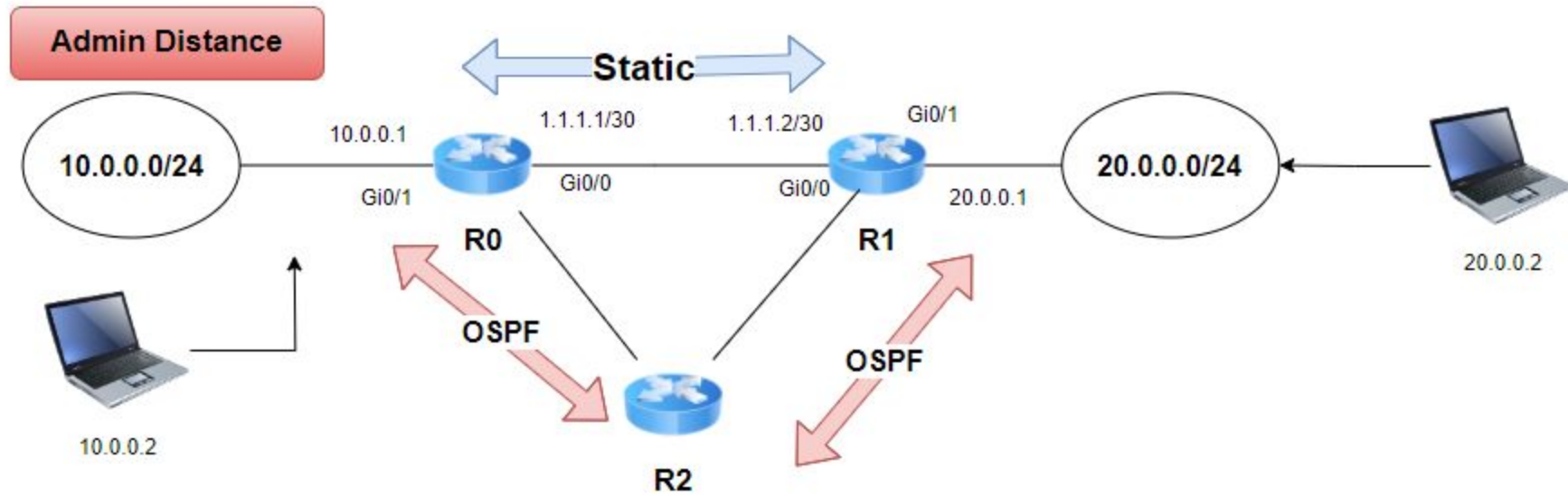
- (TTL) is a value in an Internet Protocol (IP) packet that tells network routers how many hops the packet is allowed before it must be discarded.
- Each time the packet passes through a router, the TTL decreases by 1
- When the TTL reaches 0, the packet is discarded, and the router sends an ICMP "Time Exceeded" message back to the sender.

TraceRoute

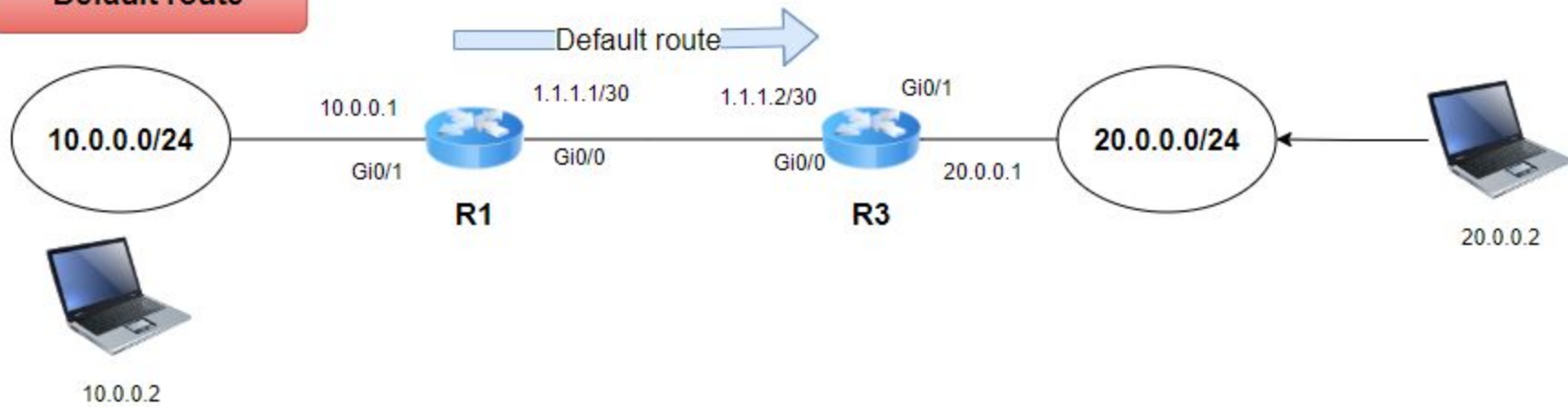
- Tool that shows the path a packet of data takes from one device to another across a network.
- Time-To-Live (TTL) values. Each time a packet reaches a router, the TTL is decreased by 1. If the TTL reaches 0, the router stops the packet and sends back an ICMP "Time Exceeded" message to the sender.

Longest Bit Match





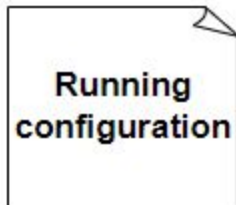
Default route



Memories



RAM



Running
configuration



Router



NVRAM



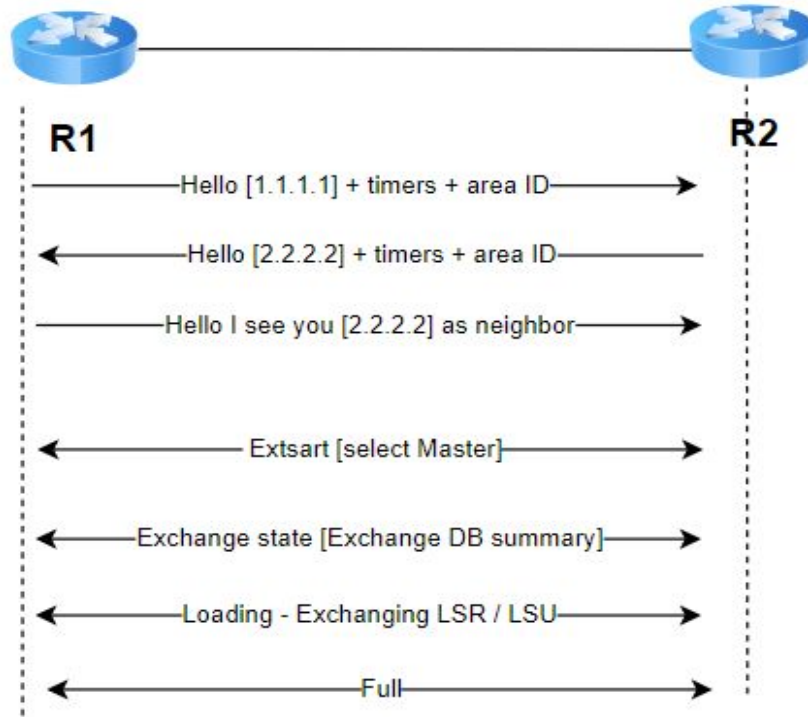
Startup
configuration

Copy Run start

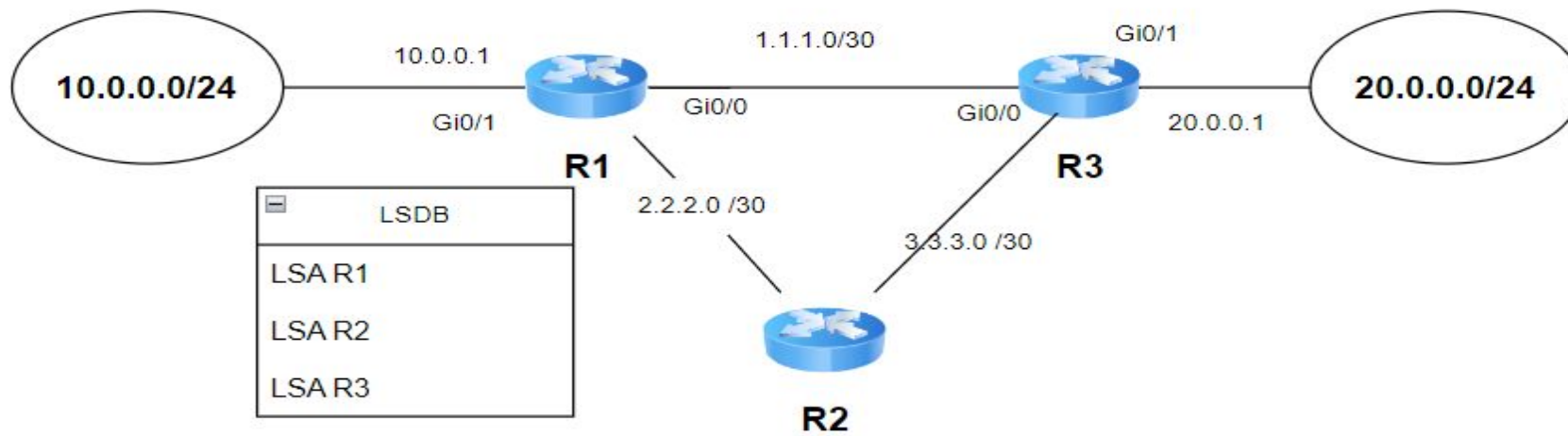
OSPF

- Hello Timer 10 seconds x Dead Timer 40 seconds (P2P)
- Hello Timer 10 seconds x Dead Timer 40 seconds (BMA)
- Admin Distance = 110
- Process ID = for multiple instance
- Area ID for DB stability

Neighborhood



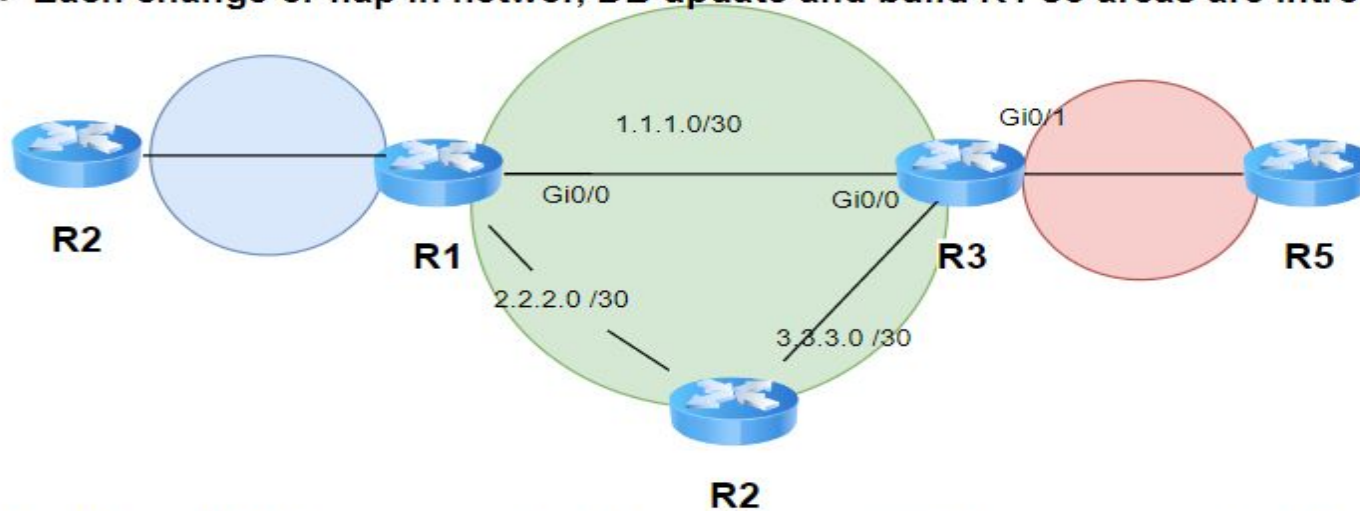
- Each router will exchange hello with info about timer + Area ID + Router ID
- Link State Advertisement (LSA) for exchange Routes
- LSA Type 1, each router will Send its own Interfaces & cost for each interface
- Each Router will take a copy of LSA and put in LSDB & then Forward LSA



- Each router will apply algorithm called Dijkstra to get best path and build RT

RT R1	
20.0.0.0 / 24	R3
3.3.3.0 /30	R2

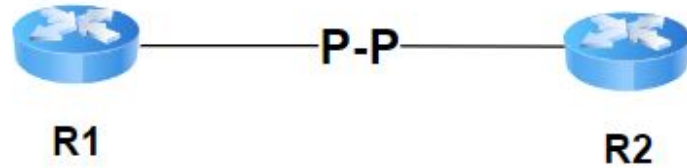
- Each change or flap in networ, DB update and build RT so areas are introduced



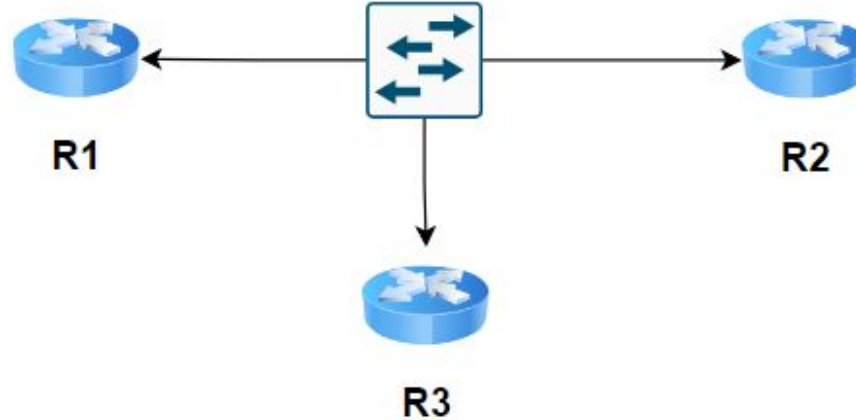
- R3 will send LSA 3 summary about networks in with R5 area to area0 routers

OSPF neighbor types

Point to Point



Broadcast Multiple Access BMA



- Elect DR to receive updates and send updates to all devices
- All devices build neighborship with DR

Designated router (DR)

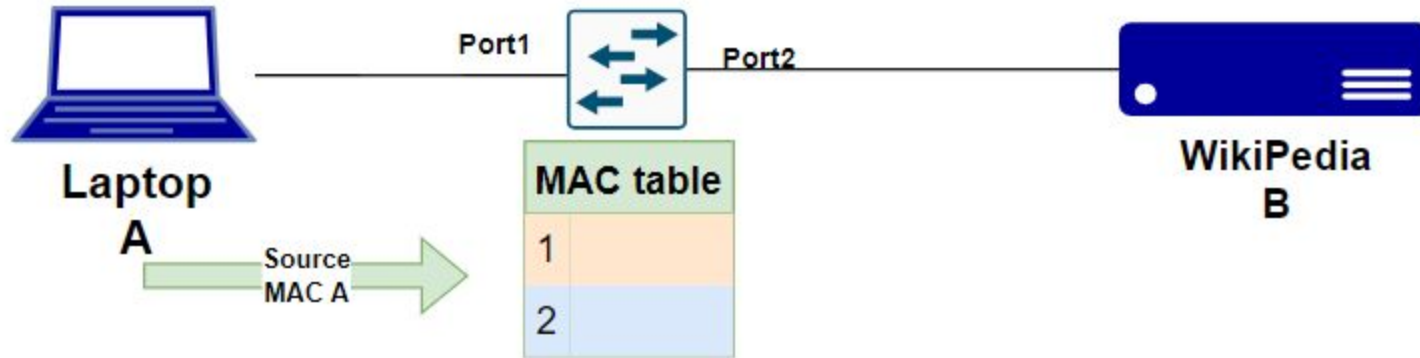
- Highest Priority
- Highest Router ID (Configuration or Highest Loopback or Highest active interface)

OSPF configuration

- (config)# Router OSPF [Process ID]
- Network area

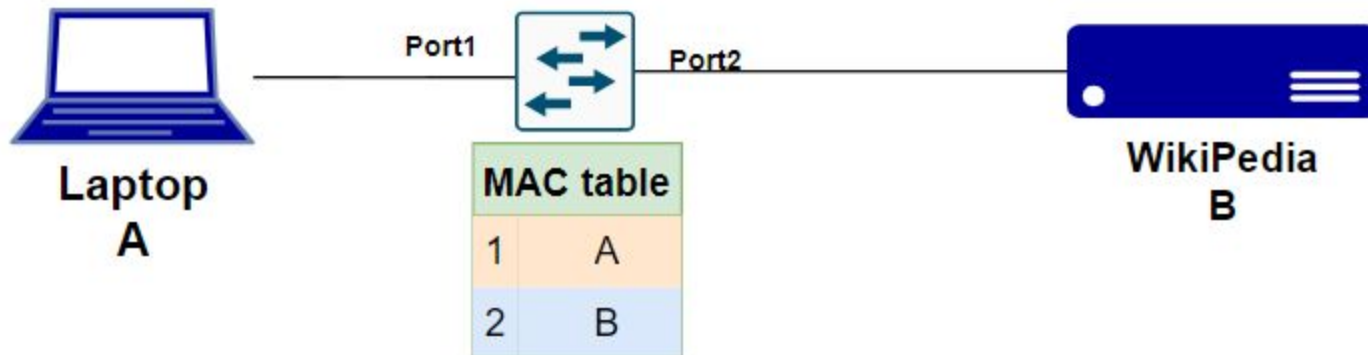
Building MAC Table

- The switch builds the MAC table by checking the **source MAC**



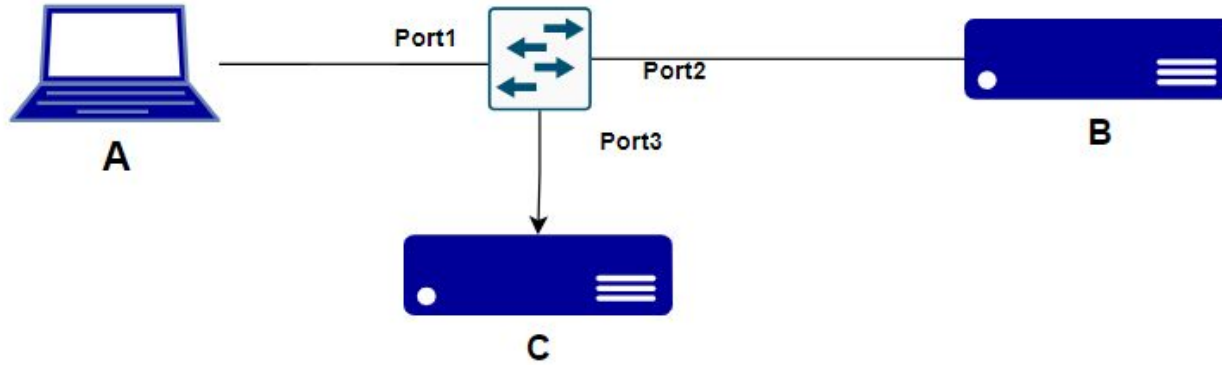
Forwarding the frames

- Forward the frames based on **DST MAC** address using MAC table



Flooding happens when

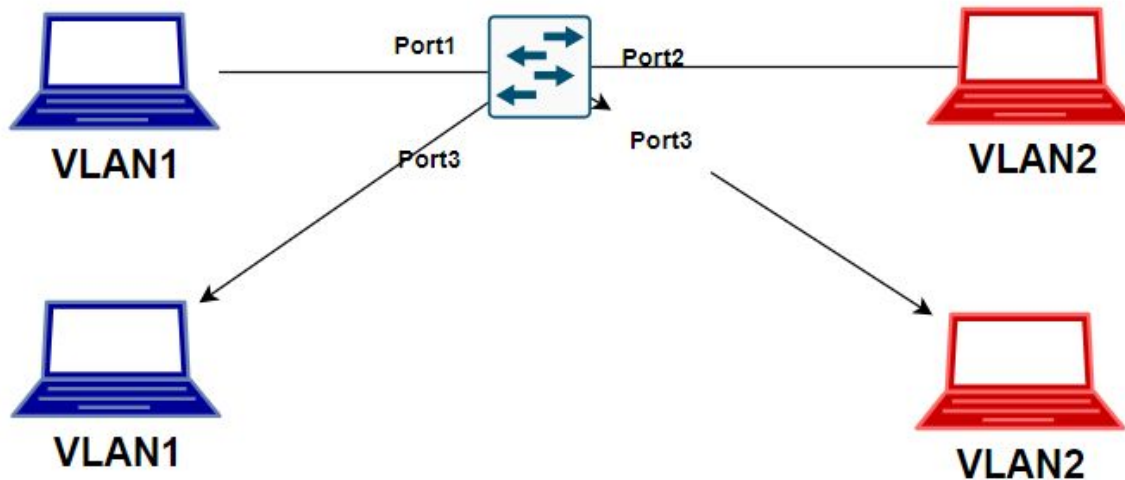
- Broadcast MAC address in frame FFFF.FFFF.FFFF
- Multicast MAC address, now can be more optimized
- Unknown unicast



~~Build MAC Table~~

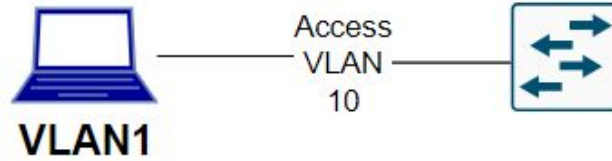
- Provide Segmentation using Separate Broadcast domain
- Flooding will happen in the ports member of the same VLAN
- Default VLAN is 1

VLAN



Access Port

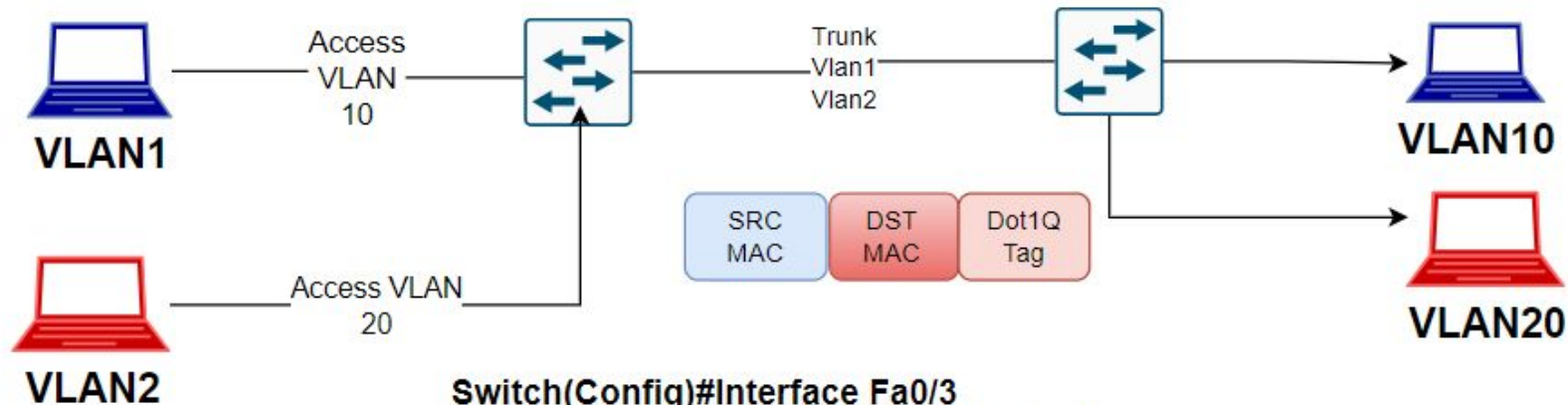
- Port is member in one VLAN
- Used to connect PCs or laptops or servers



```
Switch(Config)#Interface fa0/0  
Switch(Config)#switchport mode access  
Switch(Config)#switchport access 10
```

Trunk Port

- Member in many VLAN
- Connecting between two switches - uplink
- Tag happens using Dot1Q encapsulation



```
Switch(Config)#Interface Fa0/3  
Switch(Config)#switchport mode Trunk  
Switch(Config)#switchport Trunk allow vlan 10,20
```

Native VLAN

- Switch will forward specific VLAN untagged
- Default Native VLAN is 1

```
Switch(Config)#Interface Fa0/3
```

```
Switch(Config)#switchport mode Trunk
```

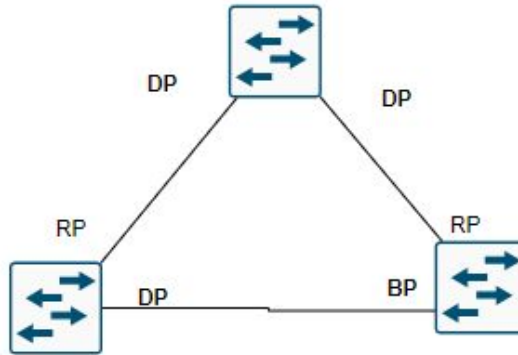
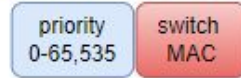
```
Switch(Config)#switchport Trunk native vlan 10
```

Spanning Tree

- Solve Broadcast Storm by blocking some ports to avoid L2 loop



- Hello BPDUs are flooded every 2 seconds
- Elect **Root switch** that has lowest switch ID



- Elect **Root Port (RP)**; best port to reach root
 1. Least cost to root based on BW
 2. least neighbor switch ID
 3. Least neighbor Port ID
- Elect **Designated Port (DP)** Best port on segment that can reach root
- All root ports are DP
 1. Least Path Cost to root
 2. Least sender SW ID
 3. Least Sender port ID
- Blocked port is not RP or DP

Status

- Discard (Listening)
- Learning (building mac table but not forwarding)
- Forwarding

Port Fast

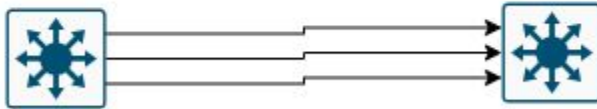
- Directly put the port in FW status
- For ports connect to PCs or servers



VLAN10

Link Aggregation Control Protocol (LACP)

- Increase link BW by bundling multiple physical ports in one logical port
- LACP used to negotiate between both SWs
- Loadbalancing by default
- All physical ports should have same config VLAN//Speed/Trunk



```
SW(Config)#Interface range Fa0/1 -3  
SW(Config)#Channel-group 1 mode active
```

```
SW(Config)#Interface ort-channel1  
SW(Config)#switchport mode trunk
```

**Cisco Discovery
Protocol (CDP)**

- Discover Neighbor device name
- my device interface

**Link Layer
Discovery protocol
LLDP**

- my neighbor device interface
- IP
- MAC

```
Switch#show lldp neighbors
```

```
Capability codes:
```

```
(R) Router, (B) Bridge, (T) Telephone, (C) DOCSIS Cable Device
```

```
(W) WLAN Access Point, (P) Repeater, (S) Station, (O) Other
```

Device ID	Local Intf	Hold-time	Capability	Port ID
Nortel IP Phone	Gi1/0/1	180	T	0019.e1e7.018d
Polycom SoundPoint IG	Gi1/0/19	180	T	0004.f22f.88b7
Baseline Switch 2426	Gi1/0/18	180	P,B	Ethernet0/26
Baseline Switch 2426	Gi1/0/22	180	P,B	Ethernet0/26

Inter VLAN Routing

- To allow communication from one VLAN to another VLAN, we have to do one two solutions

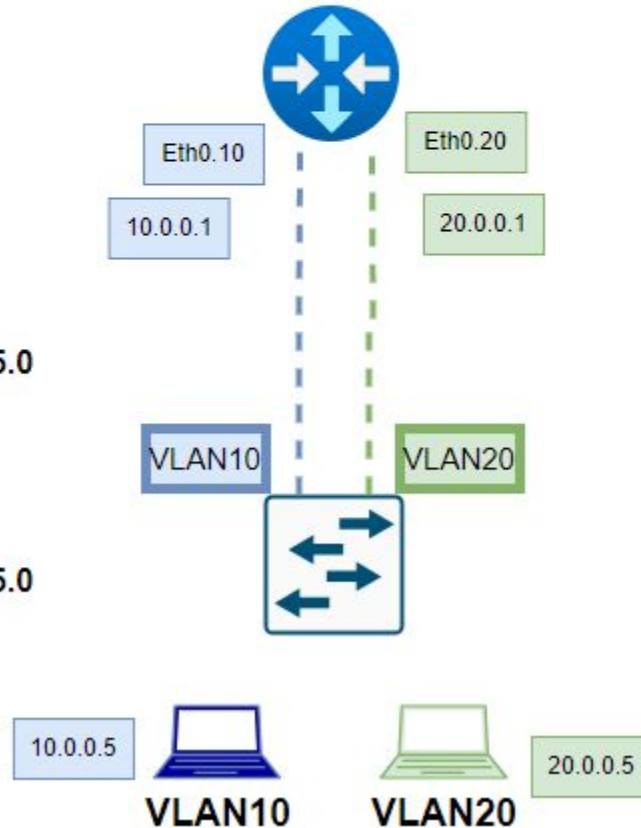
Router on Stick

- Router interface Eth0 will split to two sub interfaces
- Eth0.10 for VLAN10
- Eth0.20 for VLAN 20
- Switch port to the router will trunk carrying Vlan 10, 20

```
SW(Config)# Interface Fa0/0  
SW(Config)# switchport mode trunk
```

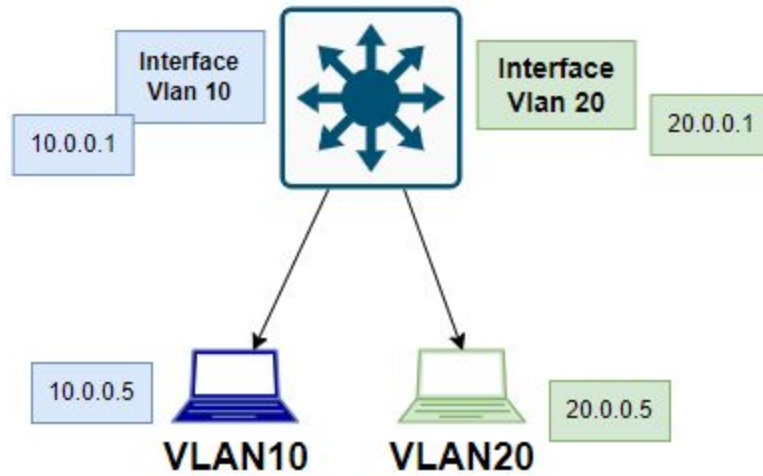
```
Router(config)# Interface Eth0.10  
Router(config)#Encapsulation dot1Q 10  
Router(config)#Ip address 10.0.0.1 255.255.255.0
```

```
Router(config)# Interface Eth0.20  
Router(config)#Encapsulation dot1Q 20  
Router(config)#Ip address 20.0.0.1 255.255.255.0
```



MultiLayer switch

- A layer3 switch can do the routing using interface VLAN configuration

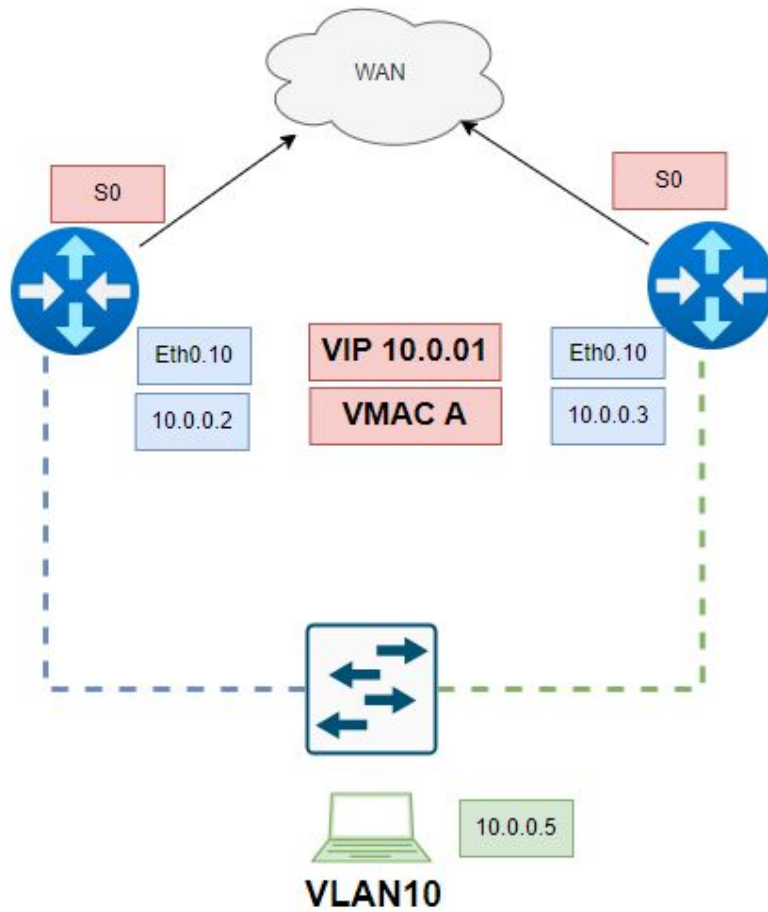


First Hop Redundancy Protocol

HSRP

VRRP

- Used to group many routers as one router
- Hello between routers on 224.0.0.2
- One router will be active the other will standby (highest priority)



```
RouterA(config)# Standby 0 ip 10.0.0.1  
RouterA(config)# Standby 0 priority 110 >> default 100  
RouterA(config)# Standby 0 preempt  
RouterA(config)# Standby 0 track s0 20
```